Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *ASAP_PLUS_PLUS* | ***Context:*** *Qatari and Arab High School Arabic Essay Writers (Grades 10–12)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark operates on English plain text with a feature pipeline (Zesch et al. attribute-independent features, Barzilay & Lapata entity grids) extracted via Stanford CoreNLP [Q31, Q32, Q33]. The deployment operates in MSA, written in right-to-left Arabic script with root-and-pattern morphology, optional diacritics (tashkeel), Arabic-specific punctuation (، ؟), and potential Gulf Arabic dialect intrusion [writing_systems, languages.dialect_influence, WEB-3, WEB-4]. Stanford CoreNLP's English pipeline cannot be applied to Arabic text directly; the canonical Arabic tools (CAMeL Tools, Camel Morph MSA, CAMeLBERT-MSA) are entirely separate ecosystems [WEB-7, WEB-8, WEB-9]. This is a complete signal-distribution mismatch, not a parameter mismatch.

ID	Evidence
Q31	"We used the attribute independent feature set provided by Zesch et al. (2015)." (p.3)
Q32	"In addition to those features, we also made use of entity grid features described in Barzilay and Lapata (2005)." (p.3)
Q33	"All the features were extracted using Stanford Core NLP (Manning et al., 2014)." (p.3)
WEB-3	QAES annotated Qatari argumentative writing with trait-specific conventions appropriate to Qatari context (https://www.mdpi.com/2306-5729/10/9/148)
WEB-4	ZAEBUC-Spoken confirms MSA/Gulf/English code-switching among Gulf students (https://arxiv.org/html/2403.18182v1)
WEB-7	CAMeL Tools provides MSA morphological analysis, POS, dialect ID and diacritization — the Arabic counterpart ecosystem to Stanford CoreNLP (https://aclanthology.org/2020.lrec-1.868.pdf)
WEB-8	Camel Morph MSA is the largest open-source MSA morphological analyzer (LREC-COLING 2024) (https://aclanthology.org/2024.lrec-main.240/)
WEB-9	CAMeLBERT-MSA is the standard pre-trained Arabic encoder used in ZAEBUC, LAILA, and TAQEEM AES systems (https://nyuad.nyu.edu/en/research/faculty-labs-and-projects/computational-approaches-to-modeling-language-lab/resources.html)

Strengths

- The structural choice to use morphology- and discourse-aware features (entity grids, coherence features) is conceptually portable to Arabic if rebuilt on Arabic-specific tools such as CAMeL Tools [Q32, WEB-7].

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Q32	"In addition to those features, we also made use of entity grid features described in Barzilay and Lapata (2005)." (p.3)
WEB-7	CAMeL Tools provides MSA morphological analysis, POS, dialect ID and diacritization — the Arabic counterpart ecosystem to Stanford CoreNLP (https://aclanthology.org/2020.lrec-1.868.pdf)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs

IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions are fundamentally different: English Latin-script LTR plain text [Q31, Q33] vs. Arabic RTL script with rich morphology and optional diacritics [writing_systems]. There is no overlap in encoding, script direction, or morphological structure.

IF-2: Qatar's digital infrastructure supports Arabic text capture and processing well — internet penetration is ~99–100% [WEB-13, WEB-14] and the Qeducation LMS supports Arabic interfaces [WEB-15, WEB-17] — but the benchmark's English-only pipeline does not connect to this infrastructure.

IF-3: Domain-specific form differences include: (a) RTL rendering and mixed-direction handling, (b) MSA/dialect code-mixing (empirically documented in Gulf student writing [WEB-3, WEB-4]), (c) diacritization decisions for literary/Quranic essays, (d) Arabic punctuation. None are addressed by the benchmark.

IF-4: Form mismatches are total: the benchmark cannot ingest Arabic input at all without a complete pipeline replacement. This represents the most severe class of external validity violation.

ID	Evidence
Q31	“We used the attribute independent feature set provided by Zesch et al. (2015).” (p.3)
Q33	“All the features were extracted using Stanford Core NLP (Manning et al., 2014).” (p.3)
WEB-13	https://datareportal.com/reports/digital-2026-qatar
WEB-14	https://thepeninsulaqatar.com/article/18/02/2025/at-100-qatar-continues-to-lead-in-global-internet-penetration
WEB-15	Qeducation LMS (CYPHER) supports Arabic-language interface — relevant target platform for natural-language Arabic feedback delivery (https://www.cypherlearning.com/blog/k-20/qatar-march-workshop-recap-empowering-students-teachers-and-parents-with-digital-technology-for-learning)
WEB-17	Official 'Qatar Education' mobile app provides RTL Arabic delivery channel (https://play.google.com/store/apps/details?id=app.qeducation.edu.gov.qa)
WEB-3	QAES annotated Qatari argumentative writing with trait-specific conventions appropriate to Qatari context (https://www.mdpi.com/2306-5729/10/9/148)
WEB-4	ZAEBUC-Spoken confirms MSA/Gulf/English code-switching among Gulf students (https://arxiv.org/html/2403.18182v1)
WEB-7	CAMeL Tools provides MSA morphological analysis, POS, dialect ID and diacritization — the Arabic counterpart ecosystem to Stanford CoreNLP (https://aclanthology.org/2020.lrec-1.868.pdf)
WEB-8	Camel Morph MSA is the largest open-source MSA morphological analyzer (LREC-COLING 2024) (https://aclanthology.org/2024.lrec-main.240/)

Information Gaps

- Whether any of the Zesch et al. attribute-independent features have validated Arabic equivalents specifically calibrated to high school student register.

Requires Expert Verification

- Choice of Arabic morphological analyzer and tokenization scheme (e.g., MADA-style, CAMeL Tools defaults) for student writing input.

Remediation

Gap 1: English Latin-script Stanford CoreNLP feature pipeline cannot ingest Arabic input.

Recommendation: Rebuild the input pipeline on CAMeL Tools / Camel Morph MSA / CAMeLBERT-MSA [WEB-7, WEB-8, WEB-9] with explicit handling for RTL rendering, optional diacritization for literary/Quranic essays, Arabic punctuation, and MSA/Gulf-Arabic dialect intrusion detection [WEB-3, WEB-4].

ID	Evidence
WEB-7	CAMeL Tools provides MSA morphological analysis, POS, dialect ID and diacritization — the Arabic counterpart ecosystem to Stanford CoreNLP (https://aclanthology.org/2020.lrec-1.868.pdf)
WEB-8	Camel Morph MSA is the largest open-source MSA morphological analyzer (LREC-COLING 2024) (https://aclanthology.org/2024.lrec-main.240/)
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